Prep sheet — Topic 1

30-second opening

"Three papers, one question: which bone can we trust when we superimpose? Doppel and Nielsen tested maxillary methods against Björk's tantalum implants; Nguyen did the same for the mandible in 3D, with bone plates standing in for implants. Register on the anterior zygomatic surface above and the chin plus inner symphysis below. Register on the palate and you lose 30 to 50 % of eruption; register on the third-molar crypt and you are off by 2 mm. All three are retrospective and selected, so they give us the direction and size of the bias, not a correction factor."

Bottom line

A palatal best fit under-reads maxillary descent (Nielsen: 1.1–2.2 mm skeletal, 1.3–1.8 mm dental over 4 y). The zygomatic surfaces drift least (Doppel: about 1 mm over 4.2 y, while ANS, A point, PNS and PTM drift 1.3–2.4 mm). In 3D the chin and inner symphysis sit within about a voxel of bone plates (0.37–0.40 mm), the crypt does not (1.94 mm), and two observers agree to 0.12 mm.

Eight numbers to remember

Number	What it is	Say it like this
30 % / 50 %	Nielsen: share of molar / incisor eruption hidden by best fit at ANS over 4 y (Tables II–IV)	"Best fit missed about a third of molar and half of incisor eruption; landmark by landmark it is 28 to 59 %."
0.5 mm	Nielsen: structural method's horizontal offset vs implants (0.53–0.66 mm, P < .01); also Doppel's tracing error	"Structural is clean vertically and half a millimetre off horizontally, which is the size of tracing error."
0.95 / 0.97 mm	Doppel Table II: anterior / posterior zygomatic surface drift	"The key ridges move about a millimetre in four years; nothing else moves less, except a palate fit the authors did not trust."
2.13 / 2.27 / 2.39 mm	Doppel Table II: A point / PNS / PTM vertical drift	"A point, PNS and PTM drift over 2 mm, and PNS was visible in only 27 of 50 films."
1.5 : 1	Doppel: orbital-floor rise to palatal-plane lowering; same as Björk & Skieller's 3 : 2 and the ABO's 3/5 : 2/5	"Same ratio, three ways of writing it. Nielsen used 1 : 1 and slightly over-read descent."
0.37 / 0.40 / 1.94 mm	Nguyen Table: chin / symphysis / third-molar crypt error vs plates	"Chin and symphysis are at about one voxel; the crypt is five times worse."
0.12 ± 0.11 mm, ICC 0.998	Nguyen aim 2: two observers on chin + symphysis, n = 25	"Below the 0.3 mm voxel. The abstract misprints it as ± 1.1."
50 / 18 / 20 + 25	Sample sizes: Doppel from ≈ 700, Nielsen from 35, Nguyen aims 1 and 2	"Fifty of seven hundred, eighteen of thirty-five, twenty plus twenty-five. All selected on image quality."

The three papers in one breath

Paper	One breath
Doppel 1994	"Fifty implant cases, 4.2 years, seventeen maxillary structures ranked by drift: key ridges best at about 1 mm; ANS, A point, PNS and PTM worst; orbital floor rises 1.5 times what the palate lowers; no inferential statistics anywhere."
Nielsen 1989	"Eighteen implant cases, 4 years, three superimpositions on the same films: best fit at ANS hides 30 to 50 % of eruption; Björk's structural method matches implants vertically and sits 0.5 mm off horizontally."
Nguyen 2018	"Twenty Class III patients with bone plates as implants, CBCT a year apart: chin 0.37, symphysis 0.40, crypt 1.94 mm; then 25 patients, two observers, 0.12 mm, ICC 0.998."

Strengths and weaknesses

Strengths

1. Implant or plate reference standards on the same images, which nobody can repeat today.
2. Within-subject method comparison (Nielsen) and a validity arm paired with a reliability arm (Nguyen).
3. The results match Björk's and Enlow's biology, so every bias has a mechanism, not just a number.

Weaknesses

1. Selection: 7 % of Doppel's records, 51 % of Nielsen's films, undeclared screening in Nguyen; all are the best-imaged cases.
2. Statistics: none in Doppel; an unnamed test, 64 uncorrected comparisons and implant rows that do not reproduce in Nielsen; an unnamed test and an implausible 0.06 mm SD in Nguyen.
3. Treatment and age mixed everywhere; a one-year interval in Nguyen; plates checked clinically and sitting next to the chin they validate.

Three questions to ask

1. "Doppel ranks structures by how far they moved, not by whether they moved consistently; the posterior key ridge moved back in 50 of 50 and is still called most reliable. Should we read 'reliable' there as 'small consistent bias'?" Why it lands: it separates bias from noise, the methodological centre of the paper.
2. "Nielsen's 30 to 50 % is four years of pubertal growth. When we superimpose a three-month progress film on the palate in a 16-year-old, how much of that bias survives?" Why it lands: it moves the paper to the chair; the honest answer is ≈ 0.1 mm or less, below tracing error.
3. "Nguyen drops the canal and the crypt that the ABO still lists. Should our 2D mandibular superimposition change, or is that a 3D-only finding?" Why it lands: it sets the board convention against the newest evidence.

Five questions you will be asked

1. Why is best fit wrong? The nasal floor resorbs and the oral palate apposes, so a palate-registered tracing rides down with the maxilla and hides its descent (Nielsen Table I: ANS down 1.62 mm on implants, up 0.28 mm on best fit).
2. What is the structural method? AP registration on the anterior surface of the zygomatic process, vertical placement with orbital-floor apposition to nasal-floor resorption 3 : 2 (Björk & Skieller; ABO 3/5 : 2/5; Doppel 1.5 : 1).
3. Which maxillary landmark drifted most in Doppel? PTM vertical at 2.39 ± 1.69 mm, then PNS 2.27 and A point 2.13; the cranial-base rows (2.5–2.6 mm) are maxillary displacement, not instability.

4. What is Nguyen's gold standard and what is wrong with it? Class III bone plates near the incisor apices, immobile clinically only, loaded by elastics, and a few millimetres from the chin they validate.
5. What does an ICC of 0.998 tell us? Near-perfect agreement between two observers' registrations; it is reproducibility, not validity, and the paper never names the variable correlated.

Exam radar

Ten facts

1. Björk 1955: tantalum implants; the reference standard.
2. Nielsen: 18 of 35, T1 10 y, 4 y; best fit misses ≈ 30 % of molar and ≈ 50 % of incisor eruption.
3. Nielsen: structural vs implant vertical NS; horizontal ≈ 0.5 mm.
4. Doppel: 50 of ≈ 700, T1 11.9 y, 4.2 y; key ridges 0.95–0.97 mm; A point 2.13, PNS 2.27, PTM vertical 2.39 mm.
5. Doppel: palatal plane vs implant line +0.92 ± 2.10° (−5° to +8°); tracing error 0.5 mm.
6. Doppel: orbital-floor rise to palatal lowering 1.5 : 1 (= Björk & Skieller 3 : 2).
7. Björk's 2D mandibular structures: anterior chin, inner symphyseal cortex, canal, third-molar germ before roots.
8. Nguyen: chin 0.37, symphysis 0.40, crypt 1.94 mm; voxel 0.3 mm; canal excluded (Krarup 2005).
9. Nguyen: n = 25, 12.6 ± 0.9 months, 0.12 ± 0.11 mm, ICC 0.998 (0.995–1.000).
10. ABO: three structural superimpositions; cranial base on the anterior sella wall, planum sphenoidale, greater wings, cribriform plate, ethmoidal crests and orbital roofs.

MCQ stems

1. Superimposing on the palatal plane registered at ANS in a growing patient will most likely: (A) over-read molar extrusion; (B) under-read molar extrusion; (C) over-read forward growth at PNS; (D) have no systematic effect. Answer B (Nielsen: ≈ 30 % under-read).
2. In Doppel 1994 the structure that moved posteriorly in all 50 cases yet was named most reliable was: (A) ANS; (B) internal palate; (C) posterior zygomatic surface; (D) orbital floor. Answer C (0.97 ± 0.43 mm; the internal palate had the same SD but was judged a poor fit).
3. In Nguyen 2018 the region with the largest error against bone plates was: (A) anterior chin; (B) inner symphysis; (C) third-molar crypt; (D) mandibular canal. Answer C (1.94 ± 0.06 mm; the canal was excluded, not measured).

Oral question

"A paper reports 0.3 mm of upper molar extrusion over three months on a palatal superimposition. How do you read it?" Model answer: "Direction first: a palatal fit under-reads descent, so the true extrusion is if anything larger, but over three months the hidden descent is ≈ 0.1 mm at Nielsen's pubertal rate (0.3–0.6 mm per year) and less at 16. The real problem is resolution: Doppel's tracing error is 0.5 mm, so 0.3 mm sits inside the noise. Direction plausible, magnitude unmeasured; ask for the method error."

Before the session

When	What
3 days before	Read the three abstracts, then Tables II–III (Doppel), I–IV (Nielsen) and the Table (Nguyen); fix the eight numbers.
2 days before	Read the Discussion sections; note the three disagreements (anterior zygomatic stability, use of the posterior zygomatic surface, 1.5 : 1 against 1 : 1).
1 day before	Rehearse the opening and the three questions aloud; check Proffit Fig 6.59 wording and the Ch 4 rotation page.
Morning of	Re-read the five expected questions; look at the two handout figures once.

Verdict in one line

Register on bone that does not remodel: the anterior zygomatic surface with the 3 : 2 rule above, chin plus inner symphysis below; everything else carries a bias these three papers have measured.